2 Goal and Area of Application

Detection of the anti-proliferative effect in cancer cell lines is often used as a marker for compounds anti-cancer effect. Due to adaptation of cancer cells to drug treatment combination therapy is applied. To evaluate the efficiency of the combined treatment in comparison to single treatment synergy detection experiments are performed.

3 Definitions and Abbreviations

SOP	Standard Operating Procedure
KI	Karolinska Institutet
CC_{50}	Half maximal cytotoxic concentration
RT	Room Temperature (18–22 °C)
FBS	Fetal Bovine Serum
PBS	Phosphate Buffered Saline
DMSO	Dimethyl Sulfoxide
ddH_2O	Double-distilled Water
CO_2	Carbon Dioxide
ATP	Adenosine Triphosphate
VPA	Valproic Acid

4 Method

4.1 General

This SOP describes the application of the CellTiter-Glo® Luminescent Cell Viability Assay (Promega) for detection of the synergetic anti-proliferative effect in pancreatic cancer cell lines based on small molecule treatment. It describes the procedure of compound- and cell-preparation to detect and analyse the anti-proliferative synergetic effect of compounds. IMPORTANT: for each new cell line and each new tested compound pair the cell number/well and the specific CC_{50}s of tested compounds should be evaluated before the application of this SOP. See also Standard Operating Procedure SOP-R4A-1.1 “Drug combination screening”.

4.2 Table 1. Information for consumables / chemicals / equipment

Product name	Supplier	Ordering no.
PANC-1	ATCC, USA	ATCC-CRL-1469
MIA PaCa-2	ATCC, USA	ATCC-CRM-CRL-1420
BxPC-3	ATCC, USA	ATCC-CRL-1687
DMEM; high glucose, no L-glutamine	Gibco, Thermo Fisher Scientific	11965092
RPMI 1640 Medium, no L-glutamine	Gibco, Thermo Fisher Scientific	31870074
PBS w/o Ca^{2+}, Mg^{2+}	VWR	392-0442

TrypLE™ Express	Gibco, Thermo Fisher Scientific	12604039
Fetal Bovine Serum, qualified, heat inactivated, Brazil	Gibco, Thermo Fisher Scientific	10500064
Penicillin 10,000 U/mL / Streptomycin 10,000 µg/mL	Gibco, Thermo Fisher Scientific	10378016
Glutamine 200mM	Gibco, Thermo Fisher Scientific	A2916801
CellTiter-Glo 2.0® Luminescent Cell Viability Assay	Promega	G9243
DMSO	Carl Roth, Germany	HN47.1
Simvastatin	Sigma Aldrich	S6196-5MG
Valproic acid	Enzo Life Sciences	ALX-550-304
384-well, Cell Culture-Treated, Flat-Bottom Microplate (Assay plate)	Corning	3764
Echo(R) Qualified 384-well Polypropylene Microplate 2.0, Clear.	Labcyte	PP-0200
384-well compound dilution plate	Greiner Bio One	784201
Echo Acoustic Liquid Handler	Beckman Coulter Life Science	Echo 650
Cell Seeding: Multidrop® Combi 384	Thermo Fisher Scientific	5840330
Ensight Multilabel Plate Reader	PerkinElmer	HH34000000
Centrifuge	Thermo Fisher Scientific	HeraeusMegafuge 16R

4.3 Procedure

1. Compound dilution preparation

As the first step in the compound dilution preparation, a dilution series of Drug 1 and Drug 2 is prepared on an Echo-compatible Labcyte source plate.
For example, in the combination of Simvastatin and Valproic acid:
Simvastatin (Drug 1) is diluted in 100% DMSO, and Valproic acid (Drug 2) is diluted in water (ddH_2O). The resulting dilution series includes seven concentration steps, plus 100% DMSO as

the final (eighth) step. The dilution series should meet the CC50 of each compound for the tested cell line.

For PANC-1 and BxPC-3, the concentration range for Simvastatin in the cell assay is 80 µM–1.25 µM, and for Valproic acid, 50–0.78125 mM with a dilution factor of 2 in each step.
For MIA PaCa-2, the concentration range for Simvastatin is 20 µM–0.3125 µM, and for Valproic acid, 50–0.78125 mM. Dilution factor: 2.
A separate control plate was prepared containing a positive (death) control (Benzethonium chloride 100 mM) and negative control (DMSO) in 16 replicates each.

Important: To avoid cytotoxic effects due to DMSO, the stock concentration of each compound should ideally be 1000 times higher than the final concentration on the cells, or at minimum high enough to ensure that the total DMSO concentration in the well does not exceed 0.5% (v/v).
As the second step, 67.5 nL of each serially diluted compound in the drug combination is transferred by acoustic dispensing using an Echo 550 to a 384-well polypropylene compound dilution plate—one dilution plate per cell assay plate—resulting in a matrix of Drug 1 and Drug 2 concentrations. The plate is then heat-sealed and used in step 3. To ensure statistical significance, the matrix should be dispensed in at least three technical replicates.

Example Drug1 on compound dilution plate

Simvastatin	Simvastatin	Simvastatin	Simvastatin	Simvastatin	Simvastatin	Simvastatin	DMSO
Simvastatin	Simvastatin	Simvastatin	Simvastatin	Simvastatin	Simvastatin	Simvastatin	DMSO
Simvastatin	Simvastatin	Simvastatin	Simvastatin	Simvastatin	Simvastatin	Simvastatin	DMSO
Simvastatin	Simvastatin	Simvastatin	Simvastatin	Simvastatin	Simvastatin	Simvastatin	DMSO
Simvastatin	Simvastatin	Simvastatin	Simvastatin	Simvastatin	Simvastatin	Simvastatin	DMSO
Simvastatin	Simvastatin	Simvastatin	Simvastatin	Simvastatin	Simvastatin	Simvastatin	DMSO
Simvastatin	Simvastatin	Simvastatin	Simvastatin	Simvastatin	Simvastatin	Simvastatin	DMSO
Simvastatin	Simvastatin	Simvastatin	Simvastatin	Simvastatin	Simvastatin	Simvastatin	DMSO

Example Drug2 on compound dilution plate

Valproic acid	Valproic acid	Valproic acid	Valproic acid	Valproic acid	Valproic acid	Valproic acid	Valproic acid
Valproic acid	Valproic acid	Valproic acid	Valproic acid	Valproic acid	Valproic acid	Valproic acid	Valproic acid
Valproic acid	Valproic acid	Valproic acid	Valproic acid	Valproic acid	Valproic acid	Valproic acid	Valproic acid
Valproic acid	Valproic acid	Valproic acid	Valproic acid	Valproic acid	Valproic acid	Valproic acid	Valproic acid
Valproic acid	Valproic acid	Valproic acid	Valproic acid	Valproic acid	Valproic acid	Valproic acid	Valproic acid
Valproic acid	Valproic acid	Valproic acid	Valproic acid	Valproic acid	Valproic acid	Valproic acid	Valproic acid
Valproic acid	Valproic acid	Valproic acid	Valproic acid	Valproic acid	Valproic acid	Valproic acid	Valproic acid
DMSO	DMSO	DMSO	DMSO	DMSO	DMSO	DMSO	DMSO

Drug1/2 combination matrix

Drug1/ Drug2							

2. Cell seeding

PANC-1 and MIA PaCa-2 are cultured in complete medium: DMEM supplemented with 10% FBS, 2 mM L-glutamine, and 1% penicillin/streptomycin. BxPC-3 are cultured in RPMI medium supplemented with 10% FBS, 2 mM L-glutamine, and 1% penicillin/streptomycin.
Cells are harvested from a T175 cm² flask at 80% confluence by washing them twice with 10 mL room temperature PBS, followed by incubation with 3 mL TrypLE™ Express for approximately 3 minutes or until detached from the flask. Cells are suspended in 10 mL of room-temperature supplemented cell culture medium, then washed by centrifugation and resuspended in 10 mL room-temperature supplemented medium to remove TrypLE™ Express.
All cell lines are diluted to a concentration of 14,286 cells per mL in supplemented cell culture medium. A volume of 35 µL of this suspension is dispensed into each well of a 384-well cell culture plate using a Multidrop Combi. The plates are left in the hood for 15 minutes to allow the cells to settle, then placed in a plastic box containing wet paper tissues. The box is covered with a lid that is placed on top but not fully tightened to allow gas exchange. The box with the cells is then incubated for 24 hours at 37 °C and 5% CO_2.

3. Compound addition

The compound dilution plate and positive control plate prepared in step 1 are diluted with 15 µL of supplemented media and then centrifuged. Then, 10 µL of the media-diluted compounds are transferred to the cell plates prepared in step 2 using the Bravo liquid handling platform equipped with a 384-well tip head. Prior to compound transfer, the diluted compounds are mixed using the pipette tips.
The plates containing compound-treated cells are placed back into the humidity box and incubated for an additional 96 hours at 37 °C and 5% CO_2.

4. Detection

The CellTiter-Glo 2.0® Luminescent Cell Viability Assay is used to detect the anti-proliferative effect of the drug combination. It is based on quantitation of ATP present in metabolically active cells.
Both the detection reagent and the plates with cells are allowed to equilibrate to room temperature. Then, 30 µL per well of the detection reagent is added to the cells using a Multidrop

Combi. The plates are placed on an orbital shaker for 5 minutes and then incubated in the dark for at least 10 minutes. The luminescence signal is detected using the Ensight microplate reader.

5. Data analysis and interpretation

Analysis of the synergetic effect is performed using the open access browser-based software "Synergyfinder": https://synergyfinder.aittokallio.group/20250729133742744077/

To prepare the data for analysis the anti-proliferative effect values obtained for each compound concentration are normalized to the value of the wells where only DMSO was added and calculated in percent %. Whereby the value of the DMSO control is set to 100 % cell viability and positive control to 0%. Then the data is ordered in an excel table to the required format for the upload to the SynergyfFinder (Table2). *Important:* concentration should be normalized to a common molar concentration. Column titles are fixed.

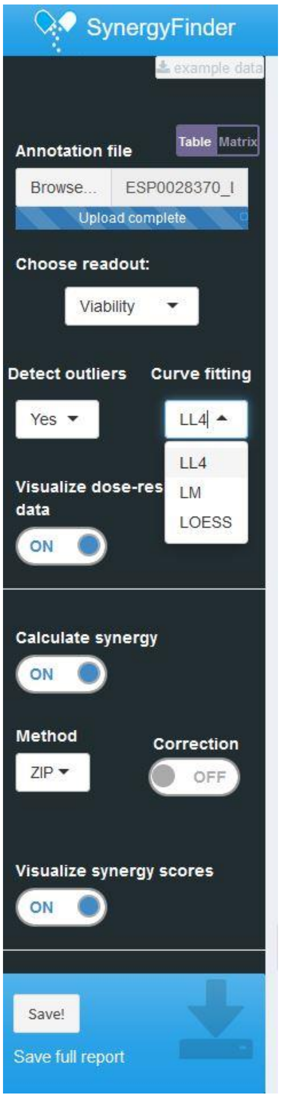
Table 2. Example table of the simvastatin/valproic acid cell viability calculation

PairIndex	Drug1	Drug2	Conc1	Conc2	Response	ConcUnit
1	Simvastatin	VPA	80000	50000000	0.17	nM
1	Simvastatin	VPA	40000	50000000	0.06	nM
1	Simvastatin	VPA	20000	50000000	0.09	nM
1	Simvastatin	VPA	10000	50000000	0.22	nM
1	Simvastatin	VPA	5000	50000000	1.94	nM
1	Simvastatin	VPA	2500	50000000	3.29	nM
1	Simvastatin	VPA	1250	50000000	0.23	nM
1	Simvastatin	VPA	0	50000000	14.45	nM
1	Simvastatin	VPA	80000	25000000	0.14	nM
1	Simvastatin	VPA	40000	25000000	0.36	nM
1	Simvastatin	VPA	20000	25000000	1.36	nM
1	Simvastatin	VPA	10000	25000000	6.45	nM
1	Simvastatin	VPA	5000	25000000	0.83	nM
1	Simvastatin	VPA	2500	25000000	11.22	nM
1	Simvastatin	VPA	1250	25000000	23.62	nM
1	Simvastatin	VPA	0	25000000	27.13	nM
1	Simvastatin	VPA	80000	12500000	0.07	nM
1	Simvastatin	VPA	40000	12500000	0.31	nM
1	Simvastatin	VPA	20000	12500000	1.96	nM
1	Simvastatin	VPA	10000	12500000	11.58	nM
1	Simvastatin	VPA	5000	12500000	6.51	nM

1	Simvastatin	VPA	2500	12500000	28.35	nM
1	Simvastatin	VPA	1250	12500000	40.97	nM
1	Simvastatin	VPA	0	12500000	27.10	nM
1	Simvastatin	VPA	80000	6250000	1.05	nM
1	Simvastatin	VPA	40000	6250000	0.48	nM
1	Simvastatin	VPA	20000	6250000	6.25	nM
1	Simvastatin	VPA	10000	6250000	8.44	nM
1	Simvastatin	VPA	5000	6250000	27.61	nM
1	Simvastatin	VPA	2500	6250000	36.68	nM
1	Simvastatin	VPA	1250	6250000	45.38	nM
1	Simvastatin	VPA	0	6250000	44.09	nM
1	Simvastatin	VPA	80000	3125000	0.19	nM
1	Simvastatin	VPA	40000	3125000	1.05	nM
1	Simvastatin	VPA	20000	3125000	3.85	nM
1	Simvastatin	VPA	10000	3125000	36.73	nM
1	Simvastatin	VPA	5000	3125000	29.66	nM
1	Simvastatin	VPA	2500	3125000	45.89	nM
1	Simvastatin	VPA	1250	3125000	49.44	nM
1	Simvastatin	VPA	0	3125000	67.21	nM
1	Simvastatin	VPA	80000	1562500	0.27	nM
1	Simvastatin	VPA	40000	1562500	0.76	nM
1	Simvastatin	VPA	20000	1562500	5.74	nM
1	Simvastatin	VPA	10000	1562500	35.19	nM
1	Simvastatin	VPA	5000	1562500	41.92	nM
1	Simvastatin	VPA	2500	1562500	94.94	nM
1	Simvastatin	VPA	1250	1562500	69.35	nM
1	Simvastatin	VPA	0	1562500	93.37	nM
1	Simvastatin	VPA	80000	781250	0.07	nM
1	Simvastatin	VPA	40000	781250	0.86	nM
1	Simvastatin	VPA	20000	781250	5.60	nM
1	Simvastatin	VPA	10000	781250	49.12	nM
1	Simvastatin	VPA	5000	781250	29.51	nM
1	Simvastatin	VPA	2500	781250	81.72	nM
1	Simvastatin	VPA	1250	781250	101.62	nM
1	Simvastatin	VPA	0	781250	84.88	nM

1	Simvastatin	VPA	80000	0	0.19	nM
1	Simvastatin	VPA	40000	0	0.48	nM
1	Simvastatin	VPA	20000	0	5.21	nM
1	Simvastatin	VPA	10000	0	27.07	nM
1	Simvastatin	VPA	5000	0	31.84	nM
1	Simvastatin	VPA	2500	0	130.80	nM
1	Simvastatin	VPA	1250	0	87.97	nM
1	Simvastatin	VPA	0	0	100.00	nM

The excel file is uploaded to the SynergyFinder. Following setting are recommended:



The calculated synergy score is a marker for a synergetic, additive or antagonistic effect of the tested compound pair. The application of scores is always content related. According to the “SynergyFinder” guide following rates are recommended:

< -10: the interaction between two drugs is likely to be antagonistic
-10 to 10: the interaction between two drugs is likely to be additive
> 10: the interaction between two drugs is likely to be synergistic.

To evaluate single concentration responses of each compound alone or with a fixed dose of thecombination pair the normalized Percent Inhibiton data was prepared according to Table 3. Shown here in Table 3 to look at VPA concentration-response curves for fixed concentrations of Simvastatin (eg S0= no Simvastatin, S20 = 20 uM Simvastatin, etc). Column Titles in the excel sheet are fixed. These were uploaded into Breeze (https://breeze.fimm.fi/v2/).

Table 3. Data upload excel example for concentration-response fitting using Breeze.

SCREEN_NAME	CONCENTRATION	DRUG_NAME	PERCENT_INHIBITION
D2_Plate1	50000	VPA_S20	98.22372119
D2_Plate1	50000	VPA_S20	99.68521641
D2_Plate1	50000	VPA_S0	98.74086565
D2_Plate1	50000	VPA_S1.3	98.21247892
D2_Plate1	50000	VPA_S2.5	98.84204609
D2_Plate1	50000	VPA_S5	97.42551996
D2_Plate1	50000	VPA_S10	95.45812254
D2_Plate1	50000	VPA_S0.3	98.6846543
D2_Plate1	50000	VPA_S0.6	96.52613828
D2_Plate1	50000	VPA_S0	87.03766161
D2_Plate1	50000	VPA_S1.3	97.76278808
D2_Plate1	50000	VPA_S2.5	98.3361439
D2_Plate1	50000	VPA_S5	98.56098932
D2_Plate1	50000	VPA_S10	95.50309162
D2_Plate1	50000	VPA_S0.3	87.58853288
D2_Plate1	50000	VPA_S0.6	98.70713884
D2_Plate1	50000	VPA_S0	83.57504216
D2_Plate1	50000	VPA_S1.3	92.55761664
D2_Plate1	50000	VPA_S2.5	98.45980888
D2_Plate1	50000	VPA_S5	95.44688027

D2_Plate1	50000	VPA_S10	99.17931422
D2_Plate1	50000	VPA_S0.3	98.93198426
D2_Plate1	50000	VPA_S0.6	98.85328836
D2_Plate1	50000	VPA_S20	99.7301855
D2_Plate1	0	VPA_S20	92.13041034
D2_Plate1	0	VPA_S20	92.92861158
D2_Plate1	0	VPA_S0	-4.78920742
D2_Plate1	0	VPA_S1.3	54.56998314
D2_Plate1	0	VPA_S2.5	82.69814503
D2_Plate1	0	VPA_S5	89.21866217
D2_Plate1	0	VPA_S10	92.05171445
D2_Plate1	0	VPA_S0.3	-5.103991006
D2_Plate1	0	VPA_S0.6	4.294547499
D2_Plate1	0	VPA_S0	3.125351321
D2_Plate1	0	VPA_S1.3	62.27093873
D2_Plate1	0	VPA_S2.5	82.33839236
D2_Plate1	0	VPA_S5	90.73636875
D2_Plate1	0	VPA_S10	92.91736931
D2_Plate1	0	VPA_S0.3	-1.866216976
D2_Plate1	0	VPA_S0.6	12.00674536
D2_Plate1	0	VPA_S0	-4.350758853
D2_Plate1	0	VPA_S1.3	56.77346824
D2_Plate1	0	VPA_S2.5	84.22709387
D2_Plate1	0	VPA_S5	89.2074199
D2_Plate1	0	VPA_S10	92.17537943
D2_Plate1	0	VPA_S0.3	-4.924114671
D2_Plate1	0	VPA_S0.6	11.54581225
D2_Plate1	0	VPA_S20	93.44575604
D2_Plate1	781.3	VPA_S20	93.11973019
D2_Plate1	781.3	VPA_S20	93.13097246
D2_Plate1	781.3	VPA_S0	7.847105115
D2_Plate1	781.3	VPA_S1.3	76.90837549
D2_Plate1	781.3	VPA_S2.5	89.66835301
D2_Plate1	781.3	VPA_S5	93.26587971
D2_Plate1	781.3	VPA_S10	92.92861158
D2_Plate1	781.3	VPA_S0.3	5.092748735

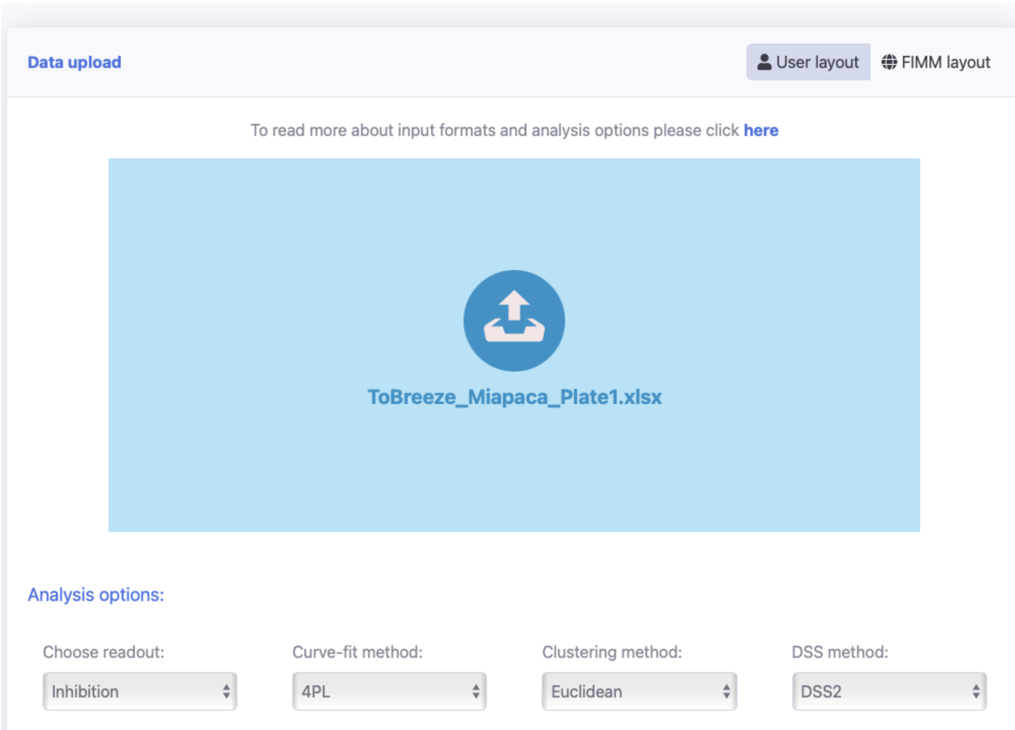
D2_Plate1	781.3	VPA_S0.6	38.87577291
D2_Plate1	781.3	VPA_S0	10.32040472
D2_Plate1	781.3	VPA_S1.3	82.48454188
D2_Plate1	781.3	VPA_S2.5	90.92748735
D2_Plate1	781.3	VPA_S5	92.25407532
D2_Plate1	781.3	VPA_S10	92.68128162
D2_Plate1	781.3	VPA_S0.3	14.40134907
D2_Plate1	781.3	VPA_S0.6	38.63968522
D2_Plate1	781.3	VPA_S0	4.440697021
D2_Plate1	781.3	VPA_S1.3	82.75435638
D2_Plate1	781.3	VPA_S2.5	91.17481731
D2_Plate1	781.3	VPA_S5	92.79370433
D2_Plate1	781.3	VPA_S10	93.56942102
D2_Plate1	781.3	VPA_S0.3	5.86846543
D2_Plate1	781.3	VPA_S0.6	38.71838111
D2_Plate1	781.3	VPA_S20	92.52388983
D2_Plate1	1562.5	VPA_S20	92.64755481
D2_Plate1	1562.5	VPA_S20	92.61382799
D2_Plate1	1562.5	VPA_S0	13.78302417
D2_Plate1	1562.5	VPA_S1.3	83.44013491
D2_Plate1	1562.5	VPA_S2.5	91.88308038
D2_Plate1	1562.5	VPA_S5	92.09668353
D2_Plate1	1562.5	VPA_S10	92.40022485
D2_Plate1	1562.5	VPA_S0.3	20.23608769
D2_Plate1	1562.5	VPA_S0.6	52.70376616
D2_Plate1	1562.5	VPA_S0	6.58797077
D2_Plate1	1562.5	VPA_S1.3	86.90275436
D2_Plate1	1562.5	VPA_S2.5	92.17537943
D2_Plate1	1562.5	VPA_S5	77.25688589
D2_Plate1	1562.5	VPA_S10	92.73749297
D2_Plate1	1562.5	VPA_S0.3	18.51602024
D2_Plate1	1562.5	VPA_S0.6	58.99943789
D2_Plate1	1562.5	VPA_S0	15.88532884
D2_Plate1	1562.5	VPA_S1.3	85.16020236
D2_Plate1	1562.5	VPA_S2.5	91.44463182
D2_Plate1	1562.5	VPA_S5	93.06351883

D2_Plate1	1562.5	VPA_S10	93.46824058
D2_Plate1	1562.5	VPA_S0.3	17.12197864
D2_Plate1	1562.5	VPA_S0.6	59.66273187
D2_Plate1	1562.5	VPA_S20	93.18718381
D2_Plate1	3125	VPA_S20	93.44575604
D2_Plate1	3125	VPA_S20	91.37717819
D2_Plate1	3125	VPA_S0	18.35862844
D2_Plate1	3125	VPA_S1.3	88.70151771
D2_Plate1	3125	VPA_S2.5	91.57953907
D2_Plate1	3125	VPA_S5	92.32152895
D2_Plate1	3125	VPA_S10	93.22091062
D2_Plate1	3125	VPA_S0.3	29.98313659
D2_Plate1	3125	VPA_S0.6	61.86621698
D2_Plate1	3125	VPA_S0	18.61720067
D2_Plate1	3125	VPA_S1.3	87.8133783
D2_Plate1	3125	VPA_S2.5	91.1410905
D2_Plate1	3125	VPA_S5	93.17594154
D2_Plate1	3125	VPA_S10	92.42270939
D2_Plate1	3125	VPA_S0.3	30.74761102
D2_Plate1	3125	VPA_S0.6	70.37661608
D2_Plate1	3125	VPA_S0	24.20460933
D2_Plate1	3125	VPA_S1.3	87.63350197
D2_Plate1	3125	VPA_S2.5	92.04047218
D2_Plate1	3125	VPA_S5	93.82799325
D2_Plate1	3125	VPA_S10	92.31028668
D2_Plate1	3125	VPA_S0.3	30.23046655
D2_Plate1	3125	VPA_S0.6	66.41933671
D2_Plate1	3125	VPA_S20	92.55761664
D2_Plate1	6250	VPA_S20	92.86115795
D2_Plate1	6250	VPA_S20	93.51320967
D2_Plate1	6250	VPA_S0	39.78639685
D2_Plate1	6250	VPA_S1.3	88.73524452
D2_Plate1	6250	VPA_S2.5	92.8836425
D2_Plate1	6250	VPA_S5	92.07419899
D2_Plate1	6250	VPA_S10	92.16413716
D2_Plate1	6250	VPA_S0.3	44.81169196

D2_Plate1	6250	VPA_S0.6	70.97245644
D2_Plate1	6250	VPA_S0	37.72906127
D2_Plate1	6250	VPA_S1.3	90.21922428
D2_Plate1	6250	VPA_S2.5	91.28724002
D2_Plate1	6250	VPA_S5	92.29904441
D2_Plate1	6250	VPA_S10	92.56885891
D2_Plate1	6250	VPA_S0.3	50.15177066
D2_Plate1	6250	VPA_S0.6	76.37998876
D2_Plate1	6250	VPA_S0	42.16975829
D2_Plate1	6250	VPA_S1.3	90.94997189
D2_Plate1	6250	VPA_S2.5	91.98426082
D2_Plate1	6250	VPA_S5	93.0747611
D2_Plate1	6250	VPA_S10	92.16413716
D2_Plate1	6250	VPA_S0.3	52.46767847
D2_Plate1	6250	VPA_S0.6	76.12141653
D2_Plate1	6250	VPA_S20	92.40022485
D2_Plate1	12500	VPA_S20	94.42383361
D2_Plate1	12500	VPA_S20	94.59246768
D2_Plate1	12500	VPA_S0	87.35244519
D2_Plate1	12500	VPA_S1.3	91.37717819
D2_Plate1	12500	VPA_S2.5	94.3901068
D2_Plate1	12500	VPA_S5	93.7717819
D2_Plate1	12500	VPA_S10	93.92917369
D2_Plate1	12500	VPA_S0.3	90.08431703
D2_Plate1	12500	VPA_S0.6	90.70264193
D2_Plate1	12500	VPA_S0	88.28555368
D2_Plate1	12500	VPA_S1.3	91.9168072
D2_Plate1	12500	VPA_S2.5	93.4232715
D2_Plate1	12500	VPA_S5	93.87296234
D2_Plate1	12500	VPA_S10	94.67116358
D2_Plate1	12500	VPA_S0.3	83.55255762
D2_Plate1	12500	VPA_S0.6	87.01517707
D2_Plate1	12500	VPA_S0	78.86453064
D2_Plate1	12500	VPA_S1.3	91.12984823
D2_Plate1	12500	VPA_S2.5	92.8049466
D2_Plate1	12500	VPA_S5	93.11973019

D2_Plate1	12500	VPA_S10	93.22091062
D2_Plate1	12500	VPA_S0.3	87.1163575
D2_Plate1	12500	VPA_S0.6	89.63462619
D2_Plate1	12500	VPA_S20	95.13209668
D2_Plate1	25000	VPA_S20	94.74985947
D2_Plate1	25000	VPA_S20	95.34569983
D2_Plate1	25000	VPA_S0	83.26025857
D2_Plate1	25000	VPA_S1.3	91.74817313
D2_Plate1	25000	VPA_S2.5	95.435638
D2_Plate1	25000	VPA_S5	93.87296234
D2_Plate1	25000	VPA_S10	95.85160202
D2_Plate1	25000	VPA_S0.3	95.45812254
D2_Plate1	25000	VPA_S0.6	95.17706577
D2_Plate1	25000	VPA_S0	95.63799888
D2_Plate1	25000	VPA_S1.3	96.76222597
D2_Plate1	25000	VPA_S2.5	97.07700956
D2_Plate1	25000	VPA_S5	97.38055087
D2_Plate1	25000	VPA_S10	96.69477234
D2_Plate1	25000	VPA_S0.3	92.32152895
D2_Plate1	25000	VPA_S0.6	93.33333333
D2_Plate1	25000	VPA_S0	85.29510961
D2_Plate1	25000	VPA_S1.3	93.02979202
D2_Plate1	25000	VPA_S2.5	93.58066329
D2_Plate1	25000	VPA_S5	92.95109612
D2_Plate1	25000	VPA_S10	93.35581788
D2_Plate1	25000	VPA_S0.3	75.7841484
D2_Plate1	25000	VPA_S0.6	84.31703204
D2_Plate1	25000	VPA_S20	95.44688027

The following settings for curve-fitting were used:



5 Further applicable documents

Manufacturer manual for CellTiter-Glo® Luminescent Cell Viability Assay (Promega): https://www.promega.com/products/cell-health-assays/cell-viability-and-cytotoxicity-assays/celltiter_glo-luminescent-cell-viability-assay/?catNum=G7570#protocols

Synergyfinder help guide: https://synergyfinder.aittokallio.group/synfin_docs/

SOP: Standard Operating Procedure SOP-R4A-1.1 "Drugs combination screening" (10.5281/zenodo.16405979)

6 Annex

No annex